SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 8

Lean Operations & Supply Chain

Lean is not a list of tools; it is a way of deciding what counts as waste. Five blocks: (1) value and the seven wastes (muda), (2) the Toyota Production System as five design choices — pull, kanban, small lots, heijunka, quality at the source — each with a precondition, (3) extending lean to the supply chain and services, (4) the Lean Startup: MVP, validated learning, Build–Measure–Learn, and (5) the KAI Logistik case: wagons returning empty ahead of Zero ODOL 2027.

Material

1 Learning objectives and roadmap

1. Define lean as **removing waste** — and value as **what the customer will pay for**.
2. Name the **seven wastes**; read Toyota as *design choices*, not a toolbox.
3. Extend lean to suppliers, warehouses, logistics, customers and services.
4. Explain the **Lean Startup**: MVP, validated learning, Build–Measure–Learn.
5. Judge a live case where lean met an organisation that had to be persuaded.

Value & Waste → Toyota System → Lean Supply Chain → Lean Startup → KAI Logistik case

Key line

Lean is not a list of tools. It is a way of deciding what counts as waste.

2 Lean glossary (to help read the slides)

Term	Meaning
Muda	Waste
Pull	Production starts only on a downstream signal
Kanban	Card authorising work; cards = inventory cap
Heijunka	Levelled volume and mix
Jidoka	Stop automatically on abnormality
Andon	Authority and signal to stop the line
Poka-yoke	Error-proofing
SMED	Cut changeover by moving setup offline
5S	Sort, set, shine, standardise, sustain
Standard work	Documented best method
Kaizen	Continuous small improvement by the people at the process
VSM	Map of value-adding vs waiting
Bullwhip	Demand swings amplify upstream
Cross-docking	Inbound goods go straight to outbound trucks
Backhaul	Loaded return trips

Supporting explanations (general lean knowledge), not slide text.

3 Lean starts with value, not cost

Customer defines value → Everything else is waste → Remove the waste → Cost falls as a result

Value is defined OUTSIDE

Only the customer can say what they will pay for.

vs

Waste is defined INSIDE

Waiting, motion, rework — seen by the people doing the work, and almost nobody else.

Start from **cost** and you cut the visible things — people, stock, suppliers — often cutting value too. Start from **value** and you know which costs carry weight.

Consequence

Lean must be run by the people at the process: **culture first, technique second**.

Value-adding	Waste (muda)
Shaping material	Waiting between steps
Assembling	Moving and handling stock
Testing to spec	Rework and inspection
Delivering	Idle inventory

<10%

Typical share of elapsed time that adds value

90%+

Waiting, moving, reworking, sitting

4 The seven wastes

Toyota's checklist. **Overproduction is the root** feeding the other six.

Waste	Meaning	Example
Overproduction (root)	Making more/earlier than demanded	Printing 1,000 brochures for 200 needed
Inventory	Stock hides every other problem	WIP-filled warehouse

Waiting	Idle time between steps	A file waits 3 days for a signature
Transport	Distance, not value	Goods shuttled between buildings
Motion	Movement that is not work	Walking far to fetch tools
Over-processing	Tighter than the customer asked	Polishing hidden parts
Defects	Rework costs capacity twice	Repairing or scrapping

≡ **Warehouse example (Ch. 14)**

Shipping defects → returns; overshipment → extra space; excess handling; waiting for parts; weak information systems.

! **Why head-office lean fails**

5 of 7 wastes are visible only to the person at the process.

5 Toyota Production System: five design choices and their preconditions

Design choice	What it means	Precondition
Pull, not push	Downstream demand authorises production; nothing is made just because capacity is free.	Fast, reliable signals
Kanban	A card authorises work; number of cards = maximum inventory.	Inventory becomes a setting
Small lots	Shorter lead time; defects surface early.	Cut setup time first — always
Heijunka	Smooth production so upstream demand stays steady.	Protects suppliers from your swings

Quality at the source	Stop defects where they are made.	Without it, pull spreads defects fast

Warning

Skip the precondition and the tool turns against you.

PUSH — a forecast authorises work

WIP piles up between steps where flow is slow.

PULL — demand authorises work

The kanban signal travels upstream. No card, no production; stock stays small.

6 The Toyota house

GOAL: best quality, lowest cost, shortest lead time — value defined by the customer

JUST-IN-TIME pillar (pull, kanban, small lots, continuous flow) | JIDOKA pillar (quality at source, stop the line, error-proofing, separate people & machines)

STABILITY: heijunka · standard work · 5S / visual management

CULTURE: kaizen · respect for people

Two messages

- **Kaizen is done by the operator, not for the operator.**
- Each pillar has a precondition: **setup reduction before small lots, stability before JIT**.

7 Sizing a kanban loop

$$k = D \times L \times (1 + S) \div C$$

Symbol	Meaning	Example
D	Demand per day	200
L	Lead time (days)	0.5
S	Safety factor	10%
C	Container size	20

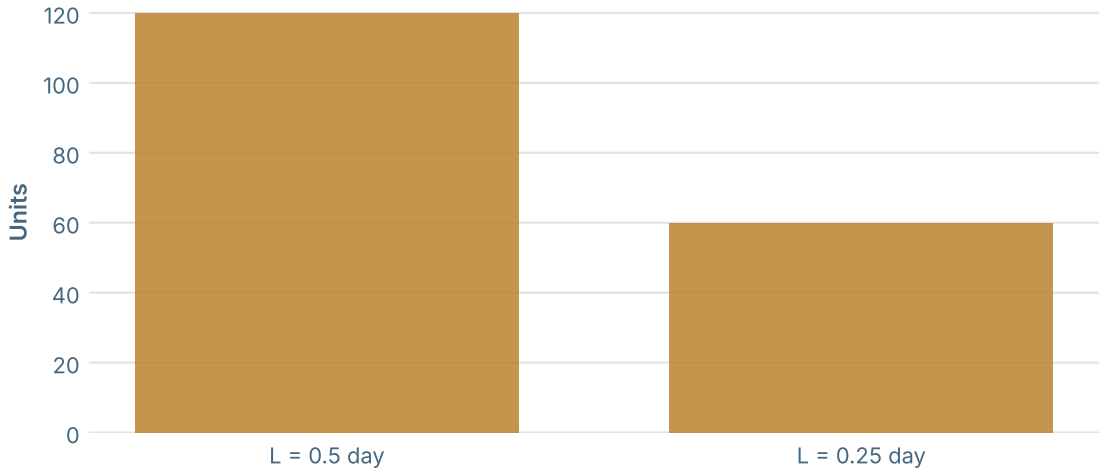
Today (L = 0.5)

k = 5.5 → **6 cards**; cap 120 units

Halve lead time (L = 0.25)

k = 2.75 → **3 cards**; cap 60 units

Kanban: lead time sets the inventory cap



Halving lead time cuts the cap from 120 to 60 units (6 → 3 cards).

The method

The card count prices your own unreliability. **Attack L, not k.** Lower k, expose a problem, fix it, lower k again. Always round up.

8 Inventory hides problems — and what lean assumes

Picture stock as the **water level** and problems as **rocks**. High stock: nothing looks wrong. Lower: one problem surfaces (supplier delay). Lean: all problems visible (supplier delay, machine breakdown) — and fixable.

> **Principle**
> Buffers don't solve problems — they hide them. Continuous improvement lowers the water deliberately.

Lean assumes	Where it breaks
Demand stable enough to level	Volatile/seasonal demand: levelling hides inventory elsewhere
Reliable, close, cooperative suppliers	Long fragile lines: no buffer + 6-week ocean leg = exposed, not lean
High quality at source	New products with unknown demand → Lean Startup

> ! **Key line**
> Removing a buffer transmits the problem unless you remove the variability too.

9 Lean tools: what each buys and what it costs

Tool	What it buys	What it costs
Value stream mapping	Splits value-adding from waiting across the flow; often <10% adds value.	Buffers vanish, problems become stoppages; improvement never ends as a project.
5S & visual management	Makes the abnormal visible at once (McDonald's, Disneyland, Speedi-Lube).	Supplier ties rebuilt, not re-priced; early gains fast, last gains cultural.

Setup reduction (SMED)	Moves setup offline; without it small lots are too costly and pull is impossible.	Front-line authority to stop the line; training time is a real budget line.

Most common failure

The wrong tool produces activity without improvement.

VSM — software release flow (Dennehy & Conboy, Dell case, Exhibit 4):

Software release value stream (days)



115 days: 75 working + 40 waiting (35% queue). After Exhibit 4, Dell case.

75 days
Value-adding

40 days
Pure waiting

115 days
Lead time — 35% queue

10 From lean plant to lean supply chain

#	Link	Lean feature	Trap
1	**Lean suppliers**	Respond to change; lower prices from removing waste	A squeezed supplier can't invest in lean
2	**Lean procurement**	Automated buying with shared visibility	Shared visibility breaks the bullwhip

3	**Lean warehousing**	Cut non-value steps from receiving to shipping	Extra stock kills efficiency
4	**Lean logistics**	Pooled orders, multistop loads, cross-docking, backhaul	Cross-docking takes storage out of the flow
5	**Lean customers**	Know their needs; value speed and flexibility	A vague customer creates upstream waste

	Inside one plant	Across a chain
Authority	Management can mandate	Every change negotiated with another P&L
Information	Everyone sees demand	Each tier sees only its next customer → bullwhip
Risk	Exposes your own problems	Exposes your partner's — they take the hit, you keep the saving
Benefit	Faster response; lower cost	Same — if the culture exists on both sides

Test
Who absorbs the variability you stop absorbing? If it's a supplier who can't say no, you only exported your stock.

11 Lean in services

Services have no goods inventory, so lean attacks **time, motion and variation**. Ch. 14's seven applications: problem-solving groups; upgraded housekeeping (McDonald's, Disneyland, Speedi-Lube); upgraded quality; clarified flows (FedEx origin-to-hub); revised equipment (Speedi-Lube drive-through pits, not hoists); levelled load (breakfast menus, take-a-number); cut unneeded steps (a hospital instrument checklist per operation).

Counterpoint

Speedi-Lube also ADDED steps that reassured customers. Lean removes waste, not reassurance.

Hospital lesson

The fix was a checklist — no new equipment, staff or budget.

12 The Lean Startup: Build–Measure–Learn

Eric Ries applies Toyota's waste logic where **demand is still a guess**.

IDEAS → BUILD → PRODUCT → MEASURE → DATA → LEARN → new ideas

Goal: **minimise time around the loop** — but what flows is knowledge.

Ch. 14

Waste = anything the customer won't pay for.

vs

Lean Startup

Waste = anything that doesn't produce **validated learning**.

So

An unwanted feature is 100% waste. Build only what the question needs; measure to decide, not to reassure.

Idea	Meaning
Falsifiable hypothesis	A belief evidence could disprove
MVP	The smallest thing that tests it
Innovation accounting	Measure learning, not activity
Validated learning	A claim backed by a real test
Persevere or pivot	The decision the evidence forces

Not a hypothesis

A hypothesis

"Users want this."

"At least 20% of trial users return within 7 days."

Lean parallel

An MVP is the small lot of product development. A pivot keeps what was learned — not a restart or an admission of failure.

13 What an MVP is and isn't — and when the Lean Startup fits

NOT an MVP

Wheel → chassis → body → car: no feedback until the end — too late.

An MVP

Skateboard (do people want to move?) → scooter (will they steer?) → bicycle (go further?) → car (scale what was validated).

Sharp definition

The least you can build that could honestly disprove your idea.

Works well	Be careful
Software & digital: build cost near zero	Capital-heavy, safety-critical: you can't MVP a subsea tool
New categories with no demand history	Regulated products: partial release may be unlawful
Corporate units testing nearby markets	"Pivot" can hide having no strategy
Decisions cheap to reverse	The loop doesn't supply judgement

Maridia

Fail-fast in 3 months vs a 7-year staged process — both right; the cost of being wrong differs.

Lean manufacturing	Lean Startup

Unknown	The process	The product itself
Waste	What the customer won't pay for	What produces no validated learning
Loop delivers	Product with less inventory	A decision: persevere or pivot
Failure	A visible, local stoppage	A disproved hypothesis — a cheap success
Use when	Demand known, problem is flow	Demand unknown, wrong build costly

14 Working through the four KAI Logistik questions

Question	Answer direction
Q1 Biggest waste	Empty return slots on trains that run anyway (overproduction and transport) and off-rail waiting; cutting waiting 20% saves ~8.7 h vs ~2.4 h for a faster train; new wagons add capacity where flow already works.
Q2 Push or pull?	The booking is the kanban card; if it arrives after Gapeka is locked, it's still push; risk moves to KALOG; shared demand visibility is missing.
Q3 Pick A, B or C	Often C as the opening of B — an MVP corridor pilot with a falsifiable target, deferring the Rp2.5T decision; watch for empty return trips not improving.
Q4 Three Board messages	Evidence of waste first, then the lever KALOG controls, then a staged ask with triggers; beware incentives that favour a big ask.

Answer directions, not an official key.

15 Six takeaways

#	Takeaway	Note

1	**Value is defined outside**	Customers buy reliable lead time, not a wagon count.
2	**Attack queue time**	Cutting waiting 20% beats a 20% faster train by ~4×.
3	**Overproduction hides waste**	A train 40% full still looks fine in a tonnage report.
4	**Removed buffers move**	The slot model shifts risk to KALOG; without visibility it's still push.
5	**Match proof to risk**	Rp2.5T fixed for decades needs every gate.
6	**Measure before deciding**	Turn Exhibit 2 into real data first.

Closing

Capacity can be bought with money. Utilisation has to be earned — and measured first.

Key takeaways

1. Lean = removing waste; value = what the customer will pay for (defined outside); waste is seen from inside the process.
2. Typically <10% of elapsed time adds value; 90%+ is waiting, moving, reworking, sitting.
3. Seven wastes: overproduction (root), inventory, waiting, transport, motion, over-processing, defects.
4. TPS = five design choices — pull, kanban, small lots (setup first), heijunka, quality at source — each with a precondition.
5. Toyota house: goal on the roof, JIT and Jidoka pillars, stability and culture foundations.
6. Kanban: k = D × L × (1 + S) ÷ C, round up; attack L, not k.
7. Inventory hides problems; removing buffers without removing variability just moves the problem.
8. Lean chain: suppliers, procurement, warehousing, logistics, customers; outside your walls change must be negotiated.
9. Lean services attack time, motion and variation; lean removes waste, not reassurance.
10. Lean Startup: waste = anything producing no validated learning; Build–Measure–Learn; MVP = least thing that could disprove the idea; persevere or pivot.

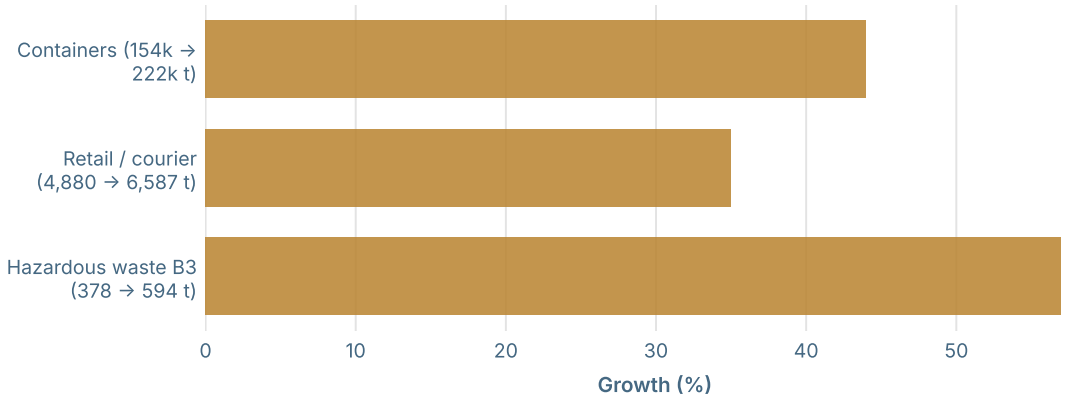
Case studies

Case & worked answers

Case C — The Wagons That Return Empty: KAI Logistik & Zero ODOL 2027

The question on Yuskal Setiawan's desk (President Director, KAI Logistik): *Should Rp2.5 trillion buy more trains — or help fill the trains we already run?* 3 weeks to prepare, 20 minutes of Board attention, Zero ODOL (no overloaded trucks) from Jan 2027, and revenue must grow +119% in 2026.

KAI Logistik new cargo growth, Jan 2025 → Jan 2026



Source: KAI Logistik case (slide 29).

	Coal	Containers
Return trip	Empty — priced into the tariff	Empty slots = pure waste
Pattern	One direction, big volume, few long contracts	Flows change with season & exports
Unloading	Tippler, whole train in hours	One box at a time (reach stacker)
Scheduling	Easy, like a production line	Many small customers, much paperwork

Most levers sit outside: track and signalling (parent KAI & regulator), maintenance (parent), partner volume (partners & customers), terminals & utilisation (KAI Logistik) — 4 of 5 outside. Capacity can be bought with money; utilisation cannot.

One container door to door, Java corridor (hours, midpoint)

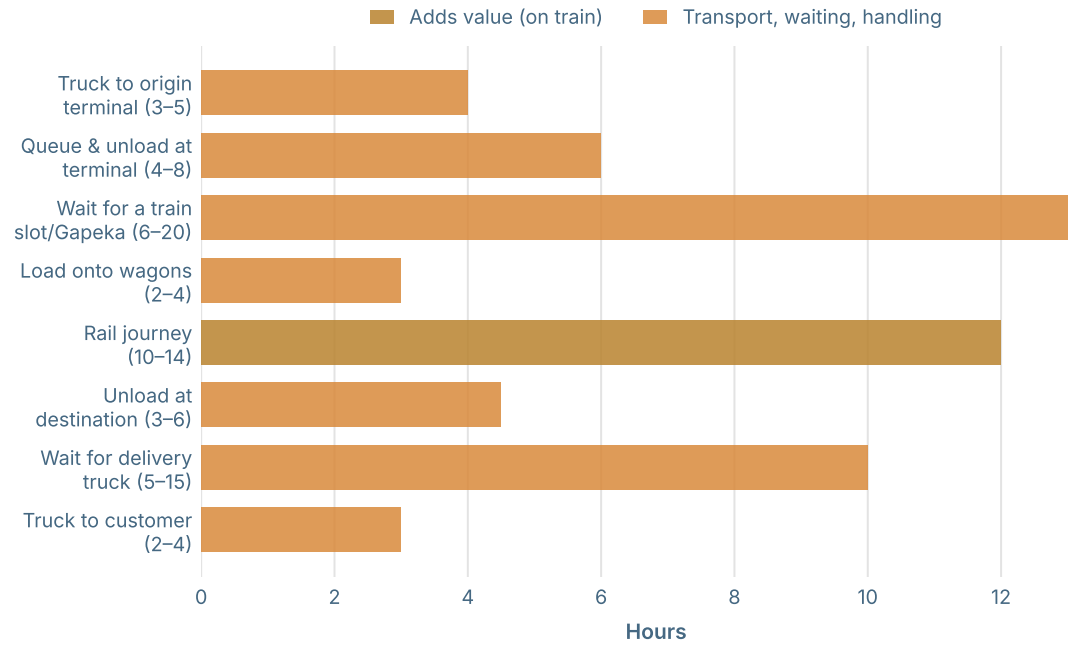


Exhibit 2 (illustrative). 35–76 h total; only 10–14 h on a moving train (18–29%).

35–76 hours door to door; only 10–14 on a moving train (≈18–29%).

	A · Capacity first	B · Utilisation first	C · One-corridor pilot
What	Full Rp2.5T: track, wagons, terminals	Slot booking, shared demand data, selling the return trip	Slot model on one Java corridor for 6 months
For	Fits national policy; fundable	Hits the biggest waste at least cost	Real data cheaply; small failure
Against	Adds capacity where flow already works	Doesn't look like "investment"	Capital decision lands after Zero ODOL

Discussion: Q1 the biggest waste and why 1,125 new flat wagons won't fix it; Q2 is the slot model real pull — what is the kanban card and what if it arrives after the Gapeka timetable is locked; Q3 pick A, B or C and name the 6-month sign you were wrong (rail had only 0.26% freight share in 2019); Q4 three Board messages, in what order.

The "≈4×" maths (Exhibit 2 midpoints)

On-train ≈ 12 h → 20% faster saves 2.4 h. Off-train ≈ 43.5 h → 20% less waiting saves 8.7 h. 8.7 ÷ 2.4 ≈ 3.6 ≈ 4×.

Practice questions

Try answering first. The answer key is below each question.

1. In lean, who defines 'value'?

A. Plant management
B. The customer — what they will pay for
C. Finance, through cost
D. Suppliers

Answer: B. Value is defined outside by the customer; waste is seen inside the process.

2. Which waste is the 'root' that feeds the other six?

A. Motion
B. Defects
C. Overproduction
D. Transport

Answer: C. Overproduction creates stock, moves and waiting, and hides defects.

3. In a pull system, what authorises production?

A. A sales forecast
B. Idle machine capacity
C. Downstream demand via a kanban signal
D. A manager's monthly target

Answer: C. The card is the authorisation; push is forecast-driven.

4. D = 200/day, L = 0.5 day, safety 10%, container 20. Number of kanban cards?

A. 5
B. 6
C. 3
D. 11

Answer: B. k = 5.5 → round up to 6 (cap 120 units).

5. Which precondition must come before small lot sizes?

A. Adding warehouse space
B. Cutting setup time (SMED)
C. Raising safety stock
D. Buying faster machines

Answer: B. Without setup reduction small lots are too costly and pull is impossible.

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6. If lead time falls from 0.5 to 0.25 day (all else equal), the card count becomes...

A. 6
B. 4
C. 3
D. 2

Answer: C. 2.75 → 3 cards, cap 60 units. Attack L, not k.

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7. The two pillars of the Toyota house are...

A. Kaizen and 5S
B. Just-in-Time and Jidoka
C. Heijunka and standard work
D. MVP and pivot

Answer: B. JIT and Jidoka are pillars; stability and culture are the foundation.

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8. The lean analogy of 'lowering the water level' means...

A. Reducing inventory so hidden problems surface and get fixed
B. Cutting headcount
C. Slowing production
D. Removing quality inspection

Answer: A. Buffers hide problems; kaizen lowers stock deliberately to expose them.

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9. A release VSM has 115 days lead time: 75 working, 40 waiting. The queue share is about...

A. 10%
B. 35%
C. 65%
D. 53%

Answer: B. 40 ÷ 115 ≈ 35%; the map shows where work waits.

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10. A firm cuts its stock by making a small supplier hold it. According to the session, this is...

A. A successful lean chain
B. Just exporting stock — variability remains
C. Heijunka
D. Cross-docking

Answer: B. If a supplier who can't say no absorbs it, you just exported your stock.

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11. Shared demand visibility across a chain mainly helps break...

A. The bullwhip effect
B. Over-processing
C. Motion
D. Setup time

Answer: A. Without it each tier sees only its next customer and swings amplify upstream.

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12. In the Lean Startup, 'waste' is...

A. Anything the customer won't pay for
B. Anything that produces no validated learning
C. Excess inventory
D. Marketing cost

Answer: B. Ries redefines waste; an unwanted feature is 100% waste.

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13. Which is a falsifiable hypothesis?

A. "Users want this app."
B. "Our product is innovative."
C. "At least 20% of trial users return within 7 days."
D. "The market is huge."

Answer: C. It must be specific and disprovable by data.

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14. Skateboard → scooter → bicycle → car illustrates...

A. Building by component
B. An MVP: each step is usable and answers a question
C. Heijunka
D. Value stream mapping

Answer: B. Wheel → chassis → body gives no feedback until the end.

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15. The Lean Startup needs most caution for...

A. Digital apps
B. New categories
C. Capital-heavy, safety-critical or regulated products
D. Easily reversed decisions

Answer: C. You can't MVP a subsea tool; partial release may be unlawful.

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16. In the KAI case, why are empty container return slots 'pure waste' but not for coal?

A. Coal is heavier
B. Coal's empty return is priced into the tariff
C. Containers need no wagons
D. Coal goes by truck

Answer: B. Coal hides the empty return in price; every empty container slot is lost money.

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17. Per Exhibit 2, why won't 1,125 new wagons fix KAI Logistik's main problem?

A. Wagons are too expensive
B. Most customer time is lost off the rails (waiting, handling)
C. Track is full
D. Customers dislike rail

Answer: B. Only 10–14 of 35–76 hours are on a moving train.

Sources

- P3MD Week 8 MK4 OSCM — Lean Operations & Supply Chain (STS)
- Ch. 14 — Lean Operations (OSCM textbook); Eric Ries — The Lean Startup
- Dennehy & Conboy — Dell case (VSM); Case C — KAI Logistik & Zero ODOL 2027